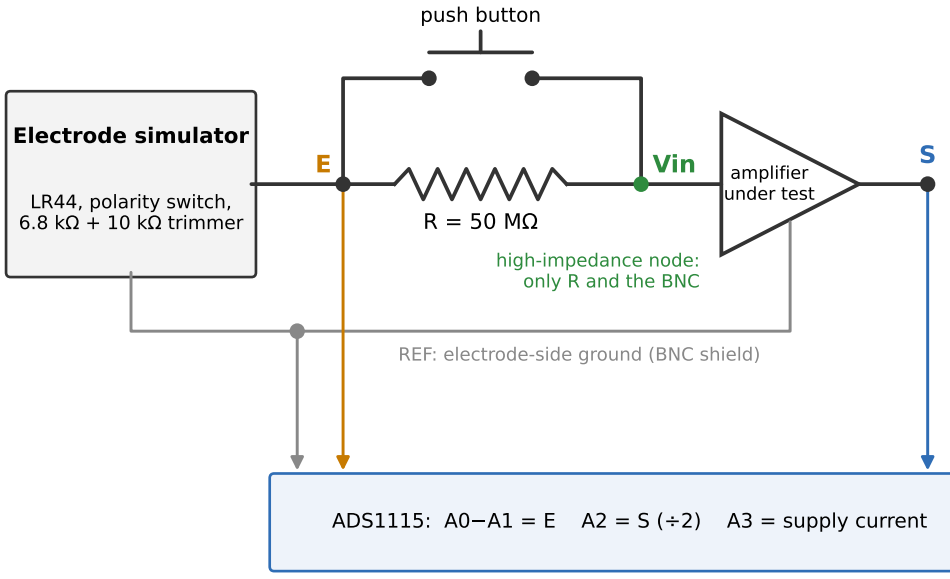

Button pressed (R shorted):

Vin = E; two points give the transfer line S(E)

Button released:

Vin from S via the line;
I = (Vin − E) / R